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The Diffusion of New Technologies

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1 Big picture

- **Question.** How do important new technologies spread across space and across skill groups, and how persistent are the local labor-market advantages enjoyed by the places where those technologies were first developed?
- **Core obstacle.** It is hard to measure “new technologies” systematically across many domains in one unified framework. Most previous work studies one technology at a time or focuses only on patents, only on jobs, or only on firms.
- **Main idea.** The paper builds a *text-to-data* pipeline that starts from patent text, filters candidate phrases using Wikipedia, then traces those technologies in online job postings and earnings calls.
- **Data innovation.** The authors combine four corpora: patent full text, Wikipedia entries, Burning Glass job postings, and earnings call transcripts.
- **Main stylized fact 1.** The origins of economically important technologies are extremely geographically concentrated. For the most economically impactful technologies, 56.3% of pioneer locations are in Silicon Valley and the Northeast Corridor.
- **Main stylized fact 2.** Technology-related jobs do spread geographically over time, but very slowly. Depending on the specification, the implied time to full diffusion is on the order of 50 or more years.
- **Main stylized fact 3.** Jobs related to new technologies are initially highly skill biased. In the year of emergence, about 57% of associated jobs require a college degree.
- **Main stylized fact 4.** Low-skill jobs spread geographically faster than high-skill jobs. Pioneer places therefore retain high-skill technology jobs for decades.
- **Mechanism emphasized by the paper.** Part of the diffusion process looks like a transition from jobs in research, development, and production (RDP) to jobs involving the use of the technology. The former remain concentrated in pioneer locations, while the latter disperse more broadly.
- **Why the paper matters.** It speaks directly to debates on place-based policy, regional inequality, technology diffusion, skill bias, and whether innovation-led growth naturally spreads beyond the handful of superstar places where frontier technologies originate.

2 Data and measurement (key objects)

2.1 Step 1: identifying novel and influential bigrams from patents

- **Patent corpus.** The starting point is the full text of U.S. patents with application years from 1976 to 2014.
- **Unit of text.** The authors extract one-word, two-word, and three-word phrases, but in the baseline they focus on **bigrams** because they are less ambiguous than unigrams and add most of the useful signal relative to trigrams.
- **Novelty filter.** They remove phrases that already appeared in pre-1970 English text from the *Corpus of Historical American English*. This isolates language associated with newer innovations.

- **Influence filter.** They keep only novel bigrams that appear in patents with at least 1,000 patent-class-and-year-normalized citations.
- **Output of step 1.** This leaves 36,563 novel and influential patent bigrams.

2.2 Step 2: identifying which candidate phrases really describe technologies

- **Wikipedia matching.** Each candidate bigram is matched to a Wikipedia page if the page title, summary, or body strongly corresponds to the bigram.
- **Why Wikipedia helps.** Pages about technologies tend to have section headings like “applications,” “uses,” “operation,” “features,” and “technical,” whereas pages about problems or management practices tend to have headings like “symptoms,” “causes,” “management,” or “risk assessment.”
- **Output of step 2.** The baseline technology list contains 4,277 technology bigrams, grouped into 2,746 Wikipedia pages. In the final paper-wide baseline, the main job-posting analysis uses 1,899 phrases grouped into 1,286 unique technology pages.
- **Interpretation.** A *technology* is a Wikipedia-title grouping, while a *technology bigram* is one of the phrases associated with that technology.

2.3 Step 3: characterizing technologies using patents and earnings calls

- **Emergence year.** For each bigram, the paper defines an emergence year as the first year in which the smoothed number of cite-weighted patent applications reaches 100 and then grows at least 10% annually for the next five years.
- **Pioneer locations.** Looking at the first 10 years after a bigram’s emergence, the authors define pioneer CBSAs as those locations that together account for at least 50% of early patenting associated with that bigram.
- **Earnings calls.** The authors search 321,373 earnings calls from 11,905 listed firms between 2002 and 2019.
- **Economically impactful technologies.** Technologies mentioned in at least 100 earnings calls are treated as especially impactful. There are 276 such technologies.
- **Why earnings calls matter.** They proxy for which technologies managers and investors view as economically important, not just scientifically novel.

2.4 Step 4: linking technologies to job postings

- **Job-posting source.** Burning Glass (BG) provides roughly 200 million postings for 2007 and 2010–2019. The baseline drops 2007 because 2008–2009 are missing.
- **Technology jobs.** The raw text of postings is used to determine whether a posting mentions a given technology bigram.
- **Coverage.** Standardized BG data provide locations and occupations for most postings, industries for a smaller subset, and employer strings for many postings.

- **Location.** Each posting is mapped to one of 917 U.S. CBSAs.
- **Occupation.** Postings are assigned six-digit SOC occupations; the paper works with 837 SOC codes after cleaning and harmonization.
- **Industry.** Industry is measured using four-digit NAICS groups.
- **Firm.** Employer strings are clustered using search results to build a firm-level panel. For firm-diffusion analysis, the sample is restricted to 10,496 larger firms posting at least one job in every sample year.

2.5 Skill measures and job-type measures

- **Skill measure.** The paper constructs a skill index for each SOC occupation using the 2015 American Community Survey (ACS): college share, postgraduate share, average wages, and years of schooling.
- **High versus low skill.** The baseline split is binary: high-skill versus low-skill occupations, though the authors show robustness to alternative definitions.
- **RDP versus use.** Using keywords plus a Word2Vec procedure trained on earnings calls, the paper classifies job postings mentioning a technology as either relating to research, development, and production (RDP) or to the use of the technology.
- **Human audit.** The human validation exercise suggests the classification is informative but noisy: the procedure correctly classifies about 63% of RDP postings and 68% of use postings in the audit sample.

2.6 Unobservables and timing

- **Technology-level age.** Different technologies are observed at different life-cycle stages in the 2010s because emergence years range from the late 1970s to the 2000s.
- **Technology-level noise.** The authors down-weight technologies with very few postings because concentration measures become noisy in thin data.
- **Main empirical timing object.** The key regressor throughout the paper is **years since emergence**. The main regressions cap this variable at 30 years.

3 Models and empirical specifications (all equations + interpretation)

3.1 Patent citation normalization

To identify influential phrases, the paper normalizes patent citations within patent class and filing year:

$$\text{NormCites}_p = \frac{\text{Citations}_p}{\text{Avg}_{\tau,t}(\text{Citations}_{p'})},$$

where patent p belongs to patent class τ and filing year t .

- **Interpretation.** This avoids treating patents in citation-heavy fields or late filing years as mechanically more important.
- **Why it matters.** The influence filter in step 1 is not based on raw citation counts, but on class-and-year-adjusted influence.

3.2 Technology-job indicator

The first core constructed variable is a posting-level indicator for whether job posting i in year t mentions technology bigram τ :

$$\text{TechnologyJob}_{i,\tau,t} = \mathbf{1}\{b_\tau \in D_{i,t}\}, \quad (1)$$

where b_τ is the technology bigram and $D_{i,t}$ is the set of bigrams contained in the job posting.

- **Implementation detail.** In the baseline, the first and last 50 words of the posting are excluded to avoid picking up boilerplate or generic firm descriptions.
- **Interpretation.** A posting is technology-related when the job itself involves using, producing, or otherwise interacting with the technology.

3.3 Normalized local share of technology jobs

To measure geographic overrepresentation of a technology in CBSA c , the paper defines

$$\text{NormalizedShare}_{c,\tau,t} = \frac{\sum_{i \in c} \text{TechnologyJob}_{i,\tau,t}}{\sum_i \text{TechnologyJob}_{i,\tau,t}} \bigg/ \frac{\#\text{Jobs}_{c,t}}{\#\text{Jobs}_t}. \quad (2)$$

- **Interpretation.** The numerator is CBSA c 's share of all postings for technology τ in year t . The denominator is CBSA c 's share of all postings in the U.S. labor market that year.
- **Meaning of the statistic.** If $\text{NormalizedShare}_{c,\tau,t} > 1$, then the technology is overrepresented in that location relative to overall job activity.
- **Why normalization matters.** It removes mechanical effects from the size of local labor markets.

3.4 Region broadening: concentration falls with age

The paper summarizes technology-level geographic concentration with the coefficient of variation across CBSAs:

$$\text{CV}_{\tau,t} = \frac{\text{sd}_c(\text{NormalizedShare}_{c,\tau,t})}{\text{mean}_c(\text{NormalizedShare}_{c,\tau,t})}.$$

It then estimates

$$\text{CV}_{\tau,t} = \alpha_0 + \beta_{RB}(t - t_{0,\tau}) + \delta_\tau + \varepsilon_{\tau,t}, \quad (3)$$

where $t_{0,\tau}$ is the emergence year of technology τ .

- **Interpretation of α_0 .** Average geographic concentration in the emergence year.
- **Interpretation of β_{RB} .** The speed at which technology jobs diffuse geographically.
- **Main estimate.** In the within-technology specification using all technologies, concentration falls by about 0.153 points of the coefficient of variation per year, implying roughly 54 years to full dispersion.
- **Economic meaning.** Technologies diffuse, but the diffusion process is very slow.

3.5 Pioneer advantage

The second core regression studies whether the places where a technology was first invented keep an employment advantage in that technology:

$$\text{NormalizedShare}_{c,\tau,t} = \alpha_0 + \beta_P \text{Pioneer}_{c,\tau} + \beta_D \text{Pioneer}_{c,\tau}(t - t_{0,\tau}) + \delta_c + \delta_\tau + \delta_t + \varepsilon_{c,\tau,t}. \quad (4)$$

- **Interpretation of β_P .** The initial pioneer premium in local technology jobs.
- **Interpretation of β_D .** The rate at which the pioneer premium erodes over time.
- **Main estimate.** For the influential-technology sample, pioneer advantage is about 108.4 percentage points in the emergence year and decays by about 3.2 percentage points per year, implying roughly 34 years to zero advantage.
- **Meaning.** Even as technologies spread, pioneer places keep a large and long-lived employment edge.

3.6 Skill measure

The average skill requirement of jobs associated with a technology bigram is constructed as

$$\text{Skill}_{\tau,t} = \frac{\sum_o N_{o,t}^\tau \chi_{o,2015}}{\sum_o N_{o,t}^\tau}, \quad (6)$$

where $N_{o,t}^\tau$ is the number of postings mentioning technology τ in occupation o and year t , and $\chi_{o,2015}$ is the skill content of occupation o measured using ACS 2015.

- **Interpretation.** Technology skill is not read directly from the posting; it is inferred from the occupational composition of the postings mentioning the technology.
- **Baseline proxy.** The main reported skill measure is the share of jobs requiring a college degree.

3.7 Skill broadening over time

The paper estimates

$$\text{Skill}_{\tau,t} = \alpha_{0,SB} + \beta_{SB}(t - t_{0,\tau}) + \delta_\tau + \varepsilon_{\tau,t}. \quad (7)$$

- **Interpretation of $\alpha_{0,SB}$.** Average skill requirement in the technology's emergence year.
- **Interpretation of β_{SB} .** The rate at which technology jobs become less skill-intensive over time.

- **Main estimate.** In the influential-technology sample, 57.1% of jobs require a college degree at emergence, and this share declines by about 0.228 percentage points per year. In the all-technology pooled specification, the decline is about 0.288 percentage points per year.
- **Meaning.** New technologies remain highly skill biased for a long time, but they do broaden gradually toward lower-skill tasks.

3.8 High-skill versus low-skill spread

The same broadening and pioneer-advantage regressions are run separately for high-skill and low-skill postings.

- **Region broadening.** In log-CV regressions, low-skill jobs spread faster than high-skill jobs.
- **Pioneer advantage.** The pioneer premium decays at about 3.2 percentage points per year for low-skill jobs and only 2.2 percentage points per year for high-skill jobs.
- **Interpretation.** The frontier places keep the design, engineering, and other high-skill work for longer, while more routine implementation and use diffuse outward faster.

3.9 RDP versus use

The same regressions are also run separately for jobs classified as RDP and jobs classified as use.

- **Region broadening.** Use jobs spread much faster than RDP jobs.
- **Pioneer advantage.** RDP jobs are initially more concentrated and their pioneer advantage erodes much more slowly.
- **Interpretation.** One mechanism behind spatial diffusion is that mature technologies move from concentrated invention/design activity to broader application in downstream occupations and places.

3.10 Extensions across firms, occupations, and industries

The paper extends NormalizedShare and CV from locations to industries, occupations, and firms.

- **Result.** Technology use diffuses across all of these dimensions, but the spread across firms and locations is more pronounced than across industries and occupations.
- **Interpretation.** New technologies diffuse across employers and places more than they completely reshuffle the sectoral and occupational map.

4 Identification and instruments (what makes the estimates credible)

4.1 This is mainly a stylized-facts paper, not a causal-treatment paper

- The paper is not built around a single natural experiment or instrumental variable.

- Instead, it constructs new objects—technology age, pioneer locations, technology-job shares, and technology skill requirements—and then studies their evolution systematically.
- The main empirical claims are therefore best read as **stylized facts disciplined by a transparent measurement strategy**, not as causal estimates of one policy or one shock.

4.2 Why the technology list is credible

- **Novelty filter.** Removing pre-1970 language avoids calling old technologies “new.”
- **Influence filter.** Requiring citation-weighted patent relevance avoids filling the sample with obscure phrases.
- **Wikipedia filter.** This separates technologies from diseases, environmental problems, and management jargon.
- **Human audit.** A manual review of sampled Wikipedia matches and sampled job postings checks that the algorithm is not obviously misclassifying large portions of the data.

4.3 Why geographic comparisons are meaningful

- **Normalization by local job share.** The main regional object compares a CBSA’s technology-job share with its overall share of U.S. jobs, so large labor markets do not mechanically appear more important.
- **Technology fixed effects.** Many core specifications compare each technology to itself over time.
- **Weighting.** Thin technology-year observations are down-weighted to reduce noise in concentration measures.

4.4 Why pioneer locations are informative

- Pioneer locations are defined from early patenting, before the paper looks at the later job-posting diffusion of that same technology.
- This helps separate where the technology was first developed from where jobs later happen to cluster.
- It also makes the “pioneer advantage” result substantively interpretable: it really is about persistent local advantage tied to early invention.

4.5 Robustness exercises

The paper shows that the main patterns are stable to:

- tighter or looser citation thresholds in the patent filter,
- adding trigrams and unigrams,
- replacing the Wikipedia screen with a human audit,

- changing the emergence-year definition,
- alternative weighting schemes,
- collapsing bigrams into Wikipedia-title technologies,
- including the 2007 BG data,
- alternative standard-error clustering strategies.
- **Bottom line.** The four headline facts are not an artifact of one fragile coding choice.

4.6 Relation to the literature

- **Technology and labor demand.** The paper connects to work on skill-biased technical change, but it adds a direct technology-age dimension and a geographic dimension.
- **Agglomeration and innovation.** It links the geography of invention to the later geography of hiring and shows that local innovation rents are very persistent.
- **Text-as-data.** Methodologically, the paper contributes a practical pipeline for turning text corpora into technology-level panels.

5 Tables and figures

5.1 Table I and Figure I: identifying technologies and emergence years

Emergence year	Wikipedia title (technology)	Technology bigrams	Number of job postings
1979	Hard disk drive	hard disk; disk drive	34,211
1980	Barcode reader	barcode reader; code reader; code scanner; barcode scanner	43,279
1981	Laser diode	emitting laser; diode laser; semiconductor laser; laser diode	7,284
1982	Personal computer	personal computer	1,752,726
1983	Flat-panel display	panel display; flat panel	27,369
1984	User interface	user interface	747,586
1985	Mobile phone	mobile telephone; cellular telephone; phones mobile; cellular phone; mobile phone; cell phone	1,832,787
1986	Facial recognition system	fit system; recognition software; recognition system; recognition technology; facial recognition	25,109
1987	Digital video	digital video	88,887
1988	Model organism	animal model	24,722
1989	Mobile device	held computer; computer device; handheld computer; mobile device	1,046,079
1990	Debit card	cards debit; card debit; debit card	260,282
1991	Flash memory	flash device; nand flash; flash memory	22,882
1992	Machine learning	learning algorithm; machine learning	491,252
1993	Financial instrument	financial instrument	43,944
1994	Active users	active user	39,671
1995	Hybrid electric vehicle	hybrid electric	8,207
1996	Digital content	digital content	144,775
1997	Multicore	multi core; core processor	29,643
1998	Information processor	data protection	176,110
1999	Unmanned aerial vehicle	aerial vehicle; unmanned aerial vehicle	24,148
2000	Transaction account	transaction account	13,012
2001	Smartphone	smart phone	910,856
2002	Online game	online game	15,254
2003	Social networking service	networking site; social networking	244,610

Emergence year	Wikipedia title (technology)	Technology bigrams	Number of job postings
2004	Electronic discovery	electronic format	56,438
2005	LED circuit	led driver	2,575
2006	Augmented reality	augmented reality	20,537
2007	Self-driving car	autonomous vehicle	18,641

Note: This table reports the top technology by number of mentions in earnings calls (in the second column) for every year of emergence between 1979 and 2007. The third column lists the associated technology bigram(s). The last column lists the number of job postings that the bigram appears in. For the year of emergence 1990, the most frequent technology is earnings calls was "software event." "No explicit software event" (because it gets dropped in our human audit) with the next most frequent technology, "unmanned aerial vehicle." See the main text for details.

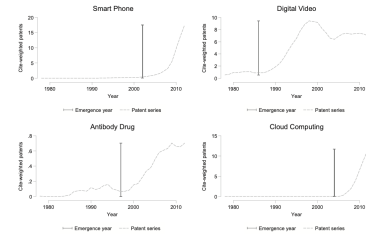


FIGURE I
Examples of Emergence Year Definition

The figure shows four examples of the attribution of emergence years. In each example, the time series plots the smoothed number of cite-weighted patents associated with the technology by year of application of the patent. For each bigram, we mark the emergence year as the first year in which (i) the technology reaches 100 cite-weighted patent applications and (ii) where the next five years had at least 10% annual growth in the (smoothed) series for each bigram. For more details, refer to text.

Read:

- Table I lists the most earnings-call-salient technology by emergence year. The examples are intuitive: personal computer, mobile phone, debit card, machine learning, hybrid electric vehicle, social networking, and self-driving car.

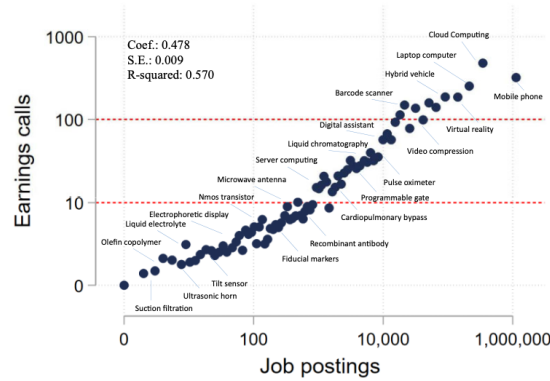


FIGURE II
Earnings Calls and Job Postings for New Technologies
The figure shows a binned scatterplot at the technology-bigram level of the number of earnings calls that mention a given technology (y-axis) against the number of job postings that mention the technology bigram (x-axis). Some examples are labeled next to their bins.

- Figure I shows how emergence years are assigned using cite-weighted patent time series.
- The point is not that the algorithm finds every technology, but that it finds a highly plausible and economically meaningful set of technologies with a disciplined dating rule.
- This figure is the foundation for the whole life-cycle analysis: once age is measured, the paper can ask how geography and skill evolve with years since emergence.

5.2 Figure II: earnings calls and job postings

Read:

- Technologies that appear more in earnings calls also appear more in job postings.
- The fitted relationship is strong: the figure reports a slope of 0.478 and an R^2 of 0.57.
- The distribution is extremely heavy tailed. The 276 technologies with more than 100 earnings-call mentions account for about 39 million job postings, or 77% of all technology-related postings.
- This figure justifies the paper’s emphasis on “economically impactful” technologies.

5.3 Table III, Figure III, and Figure IV: the concentration of invention

Read:

- Table III shows that patents mentioning new technologies are much more spatially concentrated than population or college-educated labor.
- For all new technologies, the top five CBSAs account for 33.3% of patenting. For economically impactful technologies, the top five account for 42.1%.
- Figure III shows that concentration rises monotonically with economic importance measured by earnings-call mentions.

TABLE III GEOGRAPHIC CONCENTRATION OF PATENTS, SKILL, AND EMPLOYMENT		
Total number (1)	Pct. top 5 CBSAs (2)	Top 5 CBSAs (3)
Panel A: Geographic concentration of U.S. patents		
Economically impactful	1,044,351	42.1
All new technologies	1,623,800	33.3
All patents	3,146,114	32.4
Most cited	1,044,351	32.7
Panel B: Geographic concentration of skill and employment		
College graduates	51.5 million	22.5
Employed	156.5 million	18.9
San Jose, San Francisco, New York, Seattle, Boston San Jose, San Francisco, New York, Los Angeles, Boston San Jose, New York, San Francisco, Los Angeles, Chicago San Jose, San Francisco, New York, Los Angeles, Chicago New York, Los Angeles, Chicago, Washington, San Francisco New York, Los Angeles, Chicago, Dallas, Houston		

Notes: This table reports the concentration of patents, skill, and employment across CBSAs (names shortened for presentation). San Jose: San Jose-Sunnyvale-Santa Clara, CA; San Francisco: San Francisco-Oakland-Hayward, CA; New York: New York-Newark-Jersey City, NY-NJ-PA; Seattle: Seattle-Tacoma-Bellevue, WA; Boston: Boston-Cambridge-Newton, MA-NH; Los Angeles: Los Angeles-Long Beach-Anaheim, CA; Chicago: Chicago-Naperville-Evanston, IL-IN-WI; Washington: Washington-Arlington-Alexandria, DC-VA-MD-WY; Dallas: Dallas-Fort Worth-Arlington, TX; Houston: Houston-The Woodlands-Sugar Land, TX in the United States. The measures of skill and employment are obtained from the 2015 American Community Survey. A patent is considered an economically impactful/technology patent if it mentions at least one bigram associated with an economically impactful/technology more than once. "Most cited" shows the geographic concentration of the 1,044,351 patents with the most normalized citations for comparison. This number is chosen to equal the number of patents mentioning an economically impactful technology. CBSAs in bold are those in the top five for patents that mention economically impactful technologies.

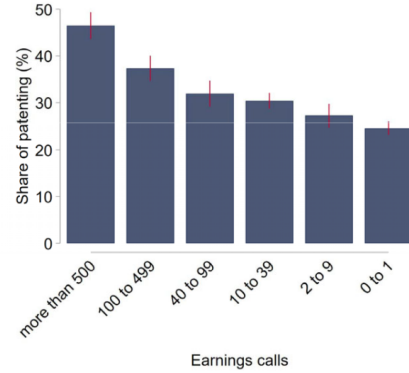


FIGURE III

Share of Patents Mentioning a New Technology Filed in Silicon Valley, New York, Seattle, and Boston, by Technology's Economic Importance (Earnings-Call Mentions)

- Figure IV is the striking map/bar-chart result: Silicon Valley accounts for 28.7% of pioneer locations, all California CBSAs for about 41.0%, and the Northeast Corridor for 27.6%, so together Silicon Valley and the Northeast Corridor host 56.3% of the pioneer locations for the most impactful technologies.

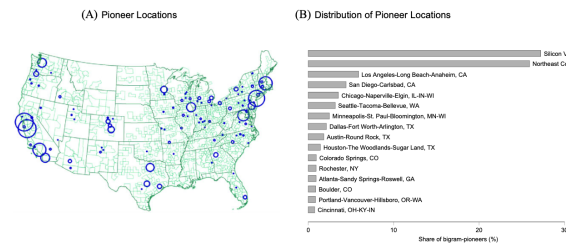


FIGURE IV

Distribution of Pioneer Locations

This figure shows the distribution of pioneer CBSAs. Panel A displays as blue circles CBSAs that are pioneer locations for at least one bigram with more than 100 earnings-call mentions. The size of the circles is proportional to the share of technology bigrams for which the CBSA is a pioneer location. Panel B shows a plot of the percentage of technology bigram-pioneer location pairs accounted for by each CBSA, for the top 20 CBSAs. We combine the CBSAs San Jose-Sunnyvale-Santa Clara, CA and San Francisco-Oakland-Hayward, CA, and label the region Silicon Valley. Similarly, we combine New York-Newark-Jersey City, Boston-Cambridge-Newton, Washington-Arlington-Alexandria, and Philadelphia-Camden-Wilmington, and label the region the Northeast Corridor.

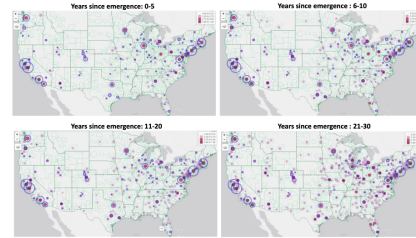


FIGURE V

Geographic Diffusion of Technology Job Postings, by Year Since Emergence

This figure plots maps with pioneer locations and job postings associated with technology bigrams by year since bigram emergence. Pioneer locations are marked with solid blue circles, and technology job postings are in solid purple circles. For each CBSA and emergence year, we calculate the share of technology bigrams for which the CBSA records a nonnegligible presence of technology jobs (*Normalized share*, $\rho \geq 10\%$) and denote a higher share of technology bigrams with a darker color. For example, the first map plots the share of technologies with a normalized share of technology job postings greater than 10% for each CBSA zero to five years since the emergence of the technology. The second map replicates this picture for 6 to 10 years after emergence, and so forth. The sample for this map only contains technologies that appear in at least 100 earnings calls.

5.4 Figure V: maps of diffusion in jobs

Read:

- The maps show both *region broadening* and *pioneer advantage* visually.
- Technology jobs do spread geographically with age.
- But the darker concentrations remain aligned with the original pioneer locations for a long time.
- The visual takeaway is that diffusion is real, but the geography of early invention leaves a very persistent footprint on the later geography of work.

5.5 Table IV and Figure VI: region broadening

Read:

TABLE IV
REGION BROADENING

	(1)	(2)	(3)
Panel A: Main specifications, Dep. Var.: <i>Coefficient of Variation</i> _{τ,t}			
Sample	EC ≥ 100	All	
<i>Years since emergence</i> _{τ,t}	−0.068*** (0.026)	−0.065*** (0.024)	−0.153*** (0.012)
Constant (CV at $t = t_{\tau,0}$)	5.577*** (0.645)	6.212*** (0.585)	8.269*** (0.271)
<i>R</i> -squared	0.019	0.013	0.825
<i>N</i>	4,270	8,347	8,347
Bigrams	428	835	835
Bigram fixed effects	No	No	Yes
Std. err. (cluster)	Wiki title	Wiki title	Wiki title
Years to zero CV	82.12	95.52	54.07
Panel B: Alternative measures of geographic concentration			
Dep. Var.:	$\frac{N_{c,t}^{Top-5}}{N_{c,t}^{All}}$	$Pct.(N_{c,t} \leq 0.1)$	$\sum_i (N_{c,t,i} - 1)^2$
<i>Years since emergence</i> _{τ,t}	−1.724*** (0.136)	−1.117*** (0.088)	−173.965** (77.247)
Constant (concentration at $t = t_{\tau,0}$)	88.781*** (3.181)	95.514*** (2.063)	13,650.448*** (1,808.008)
<i>R</i> -squared	0.846	0.922	0.679
<i>N</i>	8,347	8,347	8,347
Bigrams	835	835	835
Bigram fixed effects	Yes	Yes	Yes
Std. err. (cluster)	Wiki title	Wiki title	Wiki title
Years to zero CV	51.49	85.48	78.47

Notes. This table reports the results from regressions at the technology bigram $\times$ year level. The dependent variable is a measure of the geographic concentration of a given technology bigram's job postings in a given year. The independent variable (years since emergence) is the number of years that have elapsed since the technology's year of emergence. Panel A reports results using our baseline measure of geographic concentration: the coefficient of variation of the normalized share of a technology bigram's job postings across CBSAs. Panel B reports results using three alternative measures of geographic concentration: the mean normalized share of a technology's job postings in the top-five CBSAs relative to the mean normalized share across all CBSAs, the percentage of CBSAs with a normalized share of a technology's job postings of less than 10% (that is, the representation of CBSAs with almost no activity associated with that bigram), and the sum of squared deviations of the normalized share from one (similar to the Herfindahl-Hirschman Index). Panel A, column (1) is restricted to the sample of technology bigrams that appear in at least 100 earnings calls. The other regressions use all technology bigrams that appear in at least 1,000 job postings in our sample. Observations are weighted by the square root of the total number of job postings mentioning that technology in that year, capped at 100. The normalized share of job postings is capped at the 99th percentile of nonzero observations. Standard errors (in parentheses) are clustered by Wikipedia title (technology). All specifications indicate the fixed effects used. Years to zero CV are calculated by dividing the constant by the coefficient estimate on the years since emergence. * $p < .10$, ** $p < .05$, *** $p < .01$.

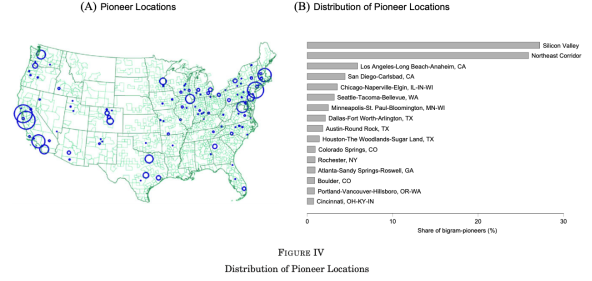


FIGURE IV
Distribution of Pioneer Locations
This figure shows the distribution of pioneer CBSAs. Panel A displays as blue circles CBSAs that are pioneer locations for at least one bigram with more than 100 earnings-call mentions. The size of the circles is proportional to the share of technology bigrams for which the CBSA is a pioneer location. Panel B shows a plot of the percentage of technology bigram-pioneer location pairs accounted for by each CBSA, for the top 20 CBSAs. We combine the CBSAs San Jose-Sunnyvale-Santa Clara, CA and San Francisco-Oakland-Hayward, CA, and label the region Silicon Valley. Similarly, we combine New York-Newark-Jersey City, Boston-Cambridge-Newton, Washington-Arlington-Alexandria, and Philadelphia-Camden-Wilmington, and label the region the Northeast Corridor.

TABLE V
PIONEER LOCATION ADVANTAGE IN TECHNOLOGY HIRING

Sample	<i>Normalized Share_{c,τ,t}</i>			
	<i>EC_τ ≥ 100</i>		All	
	(1)	(2)	(3)	(4)
<i>Pioneer_{c,τ}</i>	0.311*** (0.076)	1.084*** (0.309)	1.321*** (0.254)	1.282*** (0.243)
<i>Pioneer_{c,τ} × Years since emg_{τ,t}</i>		−0.032** (0.013)	−0.035*** (0.011)	−0.034*** (0.010)
<i>Pioneer Neighbor_{c,τ}</i>				0.158*** (0.057)
<i>Pioneer Neighbor_{c,τ} × Years since emg_{τ,t}</i>				−0.004 (0.003)
<i>R-squared</i>	0.038	0.038	0.030	0.030
<i>N</i>	3,965,122	3,965,122	7,751,024	7,751,024
<i>Bigrams</i>	428	428	835	835
<i>Bigram fixed effects</i>	Yes	Yes	Yes	Yes
<i>CBSA fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>Std. err. (cluster)</i>	Wiki title	Wiki title	Wiki title	Wiki title
<i>Rate of decline per year</i>		−0.029 (0.005)	−0.027 (0.003)	−0.026 (0.003)
<i>Implied years to zero advantage</i>		33.88	37.74	38.26

Notes. This table reports results from regressions of the *Normalized Share_{c,τ,t}* (for each CBSA × technology bigram × year) on a dummy indicating the pioneer status of the CBSA and the interaction of this dummy with the number of years that have elapsed since the bigram's emergence. The dummy variable *Pioneer Neighbor_{c,τ}* takes a value of one for non-pioneer CBSAs that are within 100 miles of the technology's pioneer locations. Columns (1) and (2) are restricted to the sample of technology bigrams that appear in at least 100 earnings calls. The other regressions use all technology bigrams that appear in at least 1,000 job postings in our sample. Observations are weighted by the square root of the total number of job postings mentioning that technology in that year, capped at 100. The normalized share of job postings is capped at the 99th percentile of nonzero observations. All specifications indicate the fixed effects used. Standard errors (in parentheses) are clustered by Wikipedia title (technology). The rate of decline per year is calculated as $\frac{\beta_D}{\beta_P}$, where β_P is the coefficient on *Pioneer_{c,τ}* and β_D is the coefficient of *Pioneer_{c,τ} × Years since emg_{τ,t}*. * $p < .10$, ** $p < .05$, *** $p < .01$.

- Table IV estimates the decline in geographic concentration over time using the coefficient of variation of normalized job shares across CBSAs.
- In the within-technology specification for all technologies, concentration falls by about 0.153 CV points per year, implying roughly 54 years to full geographic diffusion.
- Figure VI shows the same pattern nonparametrically: concentration falls materially over 30 years but remains far from zero.
- This is one of the headline facts of the paper: technologies spread, but slowly enough that local technology advantages are multi-decade phenomena.

5.6 Table V: pioneer advantage

Read:

- Pioneer places enjoy a very large employment premium in the technologies they helped invent.
- For economically impactful technologies, normalized share is about 108.4 percentage points higher in pioneer locations in the emergence year, and that advantage decays at about 3.2 percentage points annually.
- The implied time to zero pioneer advantage is about 34 years.
- So the jobs created by frontier technologies do not quickly equalize across space; the origin places keep them for a long time.

5.7 Table VI and Figure VII: high-skill versus low-skill, RDP versus use

TABLE VI
MECHANISMS: SPREAD OF HIGH-VERSUS LOW-SKILL JOBS; SPREAD OF RESEARCH, DEVELOPMENT, AND PRODUCTION JOBS VERSUS USE JOBS

Sample	All			
	High-skill job postings (1)	Low-skill job postings (2)	RDP job postings (3)	Use job postings (4)
Panel A: Region-broadening regressions				
<i>Years since emergence_{i,t}</i>	-0.027*** (0.002)	<i>log(Coefficient of Variation)_{i,t}</i> -0.038*** (0.003)	-0.014*** (0.002)	-0.036*** (0.002)
<i>R</i> -squared	0.837	0.845	0.736	0.883
<i>N</i>	8,069	8,069	6,033	6,033
Bigram fixed effects	Yes	Yes	Yes	Yes
Std. err. (cluster)	Wiki title	Wiki title	Wiki title	Wiki title
Panel B: Pioneer-advantage regressions				
<i>Pioneer_{i,t}</i>	1.319*** (0.233)	<i>Normalized Share_{i,t}</i> 1.127*** (0.255)	1.974*** (0.595)	1.581*** (0.301)
<i>Pioneer_{i,t} × Years since emg_{i,t}</i>	-0.029*** (0.010)	-0.036*** (0.010)	-0.021 (0.027)	-0.030** (0.012)
<i>R</i> -squared	0.016	0.012	0.003	0.020

TABLE VI
CONTINUED

Sample	All			
	High-skill job postings (1)	Low-skill job postings (2)	RDP job postings (3)	Use job postings (4)
<i>N</i>	8,581,946	8,395,144	5,618,723	7,769,596
Bigram fixed effects	Yes	Yes	Yes	Yes
CSRA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Wiki title	Wiki title	Wiki title	Wiki title	Wiki title
Std. err. (cluster)	-0.032 (0.004)	-0.032 (0.003)	-0.011 (0.011)	-0.019 (0.005)
Rate of decline per year				
Implied years to zero advantage	45.48	31.31	104.28	51.76

Notes: This table reports region-broadening regressions at the technology-bigram $\times$ year level (Panel A) and pioneer-advantage regressions at the technology-bigram $\times$ year $\times$ CSRA level (Panel B). Columns (1) and (2) show separate regressions for high-skill and low-skill job postings. Column (3) shows regressions for research, development, and production-related job postings (RDP), column (4) for job postings relating to the use of the technology. For definitions of these concepts, see the main text. All specifications use technology bigrams that appear in at least 100 earnings calls. Observations are weighted by the square root of the total number of job postings mentioning that technology in that year, capped at 100. All specifications indicate the fixed effects used. Standard errors (in parentheses) are clustered by Wikipedia title technology. Panel A reports results from regressions of $\log(\text{Coefficient of Variation})_{i,t}$ on the number of years since the technology's year of emergence. In a stacked specification, the difference between the coefficient on *Years since emergence_{i,t}* in columns (1) and (2) is -0.011 (std. err. = 0.003, $p = .001$). The difference between the coefficient on *Years since emergence_{i,t}* in columns (3) and (4) is -0.023 (std. err. = 0.003, $p = .000$). Both differences are thus statistically distinguishable from zero. Panel B reports results from regressions of the *Normalized share_{i,t}* for each CSRA, bigram, and year on a dummy indicating pioneer status of the CSRA and on the interaction of this dummy with the number of years that have elapsed since the bigram's emergence. The normalized share of job postings is capped at the 99th percentile of nonzero observations. In a stacked specification, the difference between estimates of the rate of decline per year in columns (1) and (2) is 0.010 (std. err. = 0.003, $p = .028$). Similarly, the difference between the rate of decline per year in columns (3) and (4) is 0.007 (std. err. = 0.007, $p = .538$). The rate of decline per year is calculated as $\frac{\partial \hat{y}_t}{\partial t}$, where $\hat{y}_t$ is the coefficient on *Pioneer_{i,t}* and $\hat{y}_t$ is the coefficient of *Pioneer_{i,t} × Years since emg_{i,t}*. * $p < .10$, ** $p < .05$, *** $p < .01$.

Read:

- Low-skill jobs spread geographically faster than high-skill jobs.
- High-skill pioneer advantage lasts longer: about 45 years to zero versus 31 years for low-skill jobs.
- Use jobs spread much faster than RDP jobs, and pioneer advantage in RDP jobs is far more persistent.
- This is the paper's main mechanism result: technology adoption diffuses outward through use, while the frontier RDP activities remain concentrated in the pioneering places.

5.8 Table VII and Figure VIII: skill broadening

Read:

- In the emergence year, about 57.1% of jobs associated with a technology require a college degree.
- The college-requirement share falls by about 0.228 percentage points per year in the influential-technology sample and 0.288 percentage points per year in the main all-technology pooled specification.
- Even 30 years later, the share is still around 50.2%, far above the population benchmark.
- The message is not that technologies quickly democratize opportunity. They do broaden, but they remain unusually skill intensive for a very long time.

5.9 Table VIII: mechanisms behind skill broadening

Read:

- The decline in skill requirements is partly explained by the falling share of RDP jobs and the rising share of use jobs as technologies mature.
- Controlling for the share of RDP jobs attenuates the skill-broadening coefficient by roughly 22%.
- By contrast, adding controls for training or experience requirements does not attenuate the skill-broadening result.

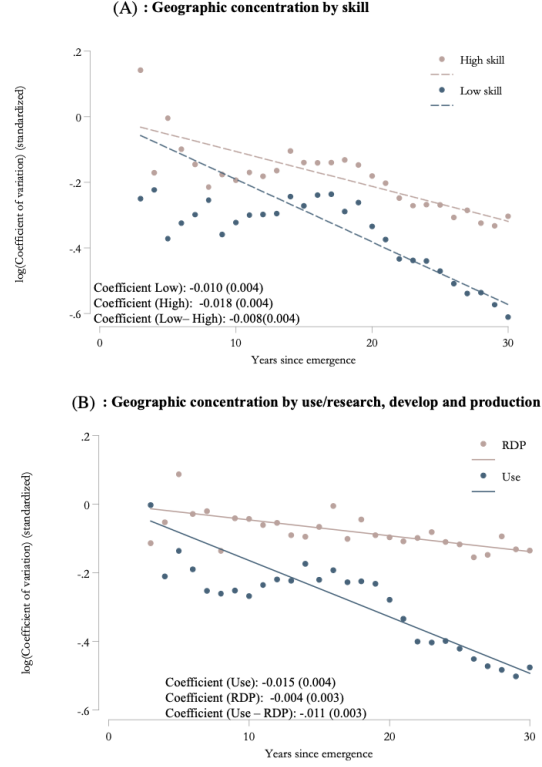


FIGURE VII
Geographic Concentration Relative to Year Since Emergence, by Skill and Type of Job Posting

This figure plots a binned scatterplot at the technology-bigram $\times$ year $\times$ skill-category level of the log of the coefficient of variation (CV) of the normalized share of bigram job postings relative to the year since emergence, by skill level of job posting (high and low). The CV is calculated as the ratio of the standard deviation to the mean of the normalized share across CBSAs for each technology-bigram $\times$ year $\times$ skill-category triplet. The red (pale gray) dots represent high-skill postings, and the blue (darker gray) dots represent low-skill postings. The fitted lines weigh observations by the square root of the total number of postings for a technology bigram in a year, capped at 100.

TABLE VII
SKILL BROADENING

Panel A: Main specification, Dep. Var.: <i>Share College Educated</i> _{<i>t,t</i>} × 100				
Sample	EC ≥ 100	All		
	(1)	(2)	(3)	(4)
Constant (share coll. ed. at <i>t</i> = <i>t</i> _{<i>t,0</i>})	57.078*** (2.135)	59.095*** (1.794)	57.475*** (2.294)	63.898*** (0.877)
<i>Years since emergence</i> _{<i>t,t</i>}	−0.228** (0.092)	−0.288*** (0.079)	−0.218*** (0.100)	−0.493*** (0.038)
<i>R</i> -squared	0.017	0.019	0.024	0.910
<i>N</i>	4,270	8,347	8,347	8,347
Bigrams	428	835	835	835
Year fixed effects	No	No	Yes	No
Bigram fixed effects	No	No	No	Yes
Standard errors (cluster)	Wiki title	Wiki title	Wiki title	Wiki title
Implied years to average skill	117.23	100.03	124.37	68.08

Panel B: Alternative measures of skill			
Dep. Var	<i>Years of Schooling</i> _{<i>t,t</i>}	<i>Share Post Graduates</i> _{<i>t,t</i>} × 100	<i>Average Wage</i> _{<i>t,t</i>}
	(1)	(2)	(3)
Constant (skill at <i>t</i> = <i>t</i> _{<i>t,0</i>})	15.504*** (0.049)	22.617*** (0.476)	71,521.317*** (877.685)
<i>Years since emergence</i> _{<i>t,t</i>}	−0.024*** (0.002)	−0.149*** (0.020)	−505.134*** (37.575)
<i>R</i> -squared	0.915	0.905	0.889
<i>N</i>	8,347	8,347	8,347
Bigrams	835	835	835
Bigram fixed effects	Yes	Yes	Yes
Std. err. (cluster)	Wiki title	Wiki title	Wiki title
Implied years to average skill	77.59	78.03	69.72

Notes. This table reports results from regressions at the technology-bigram $\times$ year level. The dependent variable is a measure of the average skill requirement of a technology bigram's job postings in a given year. The independent variable is the number of years that have elapsed since the technology's emergence. The dependent variable in Panel A is the average share of job postings mentioning technology bigram τ in year t that require a college degree. Panel B shows results corresponding to Panel A, column (4) for alternative measures of skill associated with technology-bigram job postings: average years of schooling (column (1)), share of postgraduates (column (2)), and average wage (column (3)). Panel A, column (1) is restricted to the sample of technology bigrams that appear in at least 100 earnings calls. The other regressions use all technology bigrams that appear in at least 1,000 job postings in our sample. Observations are weighted by the square root of the total number of job postings mentioning that technology in that year, capped at 100. All specifications indicate the fixed effects used. Standard errors (in parentheses) are clustered by Wikipedia title (technology). "Implied years to average skill" is determined by $-\frac{(\text{Constant} - \text{Average Population Skill})}{\beta_{\text{YSE}}(\text{Years since emergence}_{t,t})}$, where $\text{Average Population Skill}$ represents the weighted average skill of the U.S. population according to the 2015 ACS Survey. * $p < .10$, ** $p < .05$, *** $p < .01$.

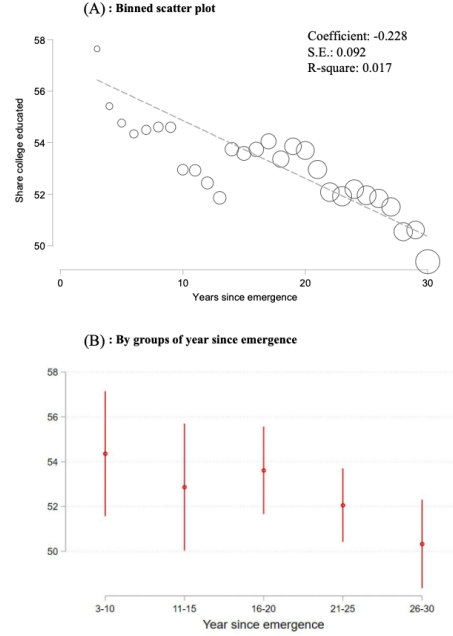


FIGURE VIII

Share of Technology Job Postings Requiring a College Education, by Year Since Emergence

Panel A shows a binned scatterplot at the technology-bigram $\times$ year level of the share of technology postings requiring a college education by year since emergence. The share of college-educated postings for each technology-bigram $\times$ year observation is measured as discussed in Section V. Each dot represents a weighted average over technology bigrams of the share for each year since emergence, where the weight is the square root of the number of postings for a bigram in a year, capped at 100. The circle sizes are proportional to the same weight. The regression line in the plot corresponds to a regression of the share of college-educated postings on the year since emergence as in Table VII, Panel A, column (1). Panel B estimates the average of the of the same dependent variable separately for five groups (3–10, 11–15, 16–20, 21–25, and 26–30+), also showing standard error bands for these estimates. Year since emergence is capped at 30 years. We only include technology bigrams that appear in at least 100 earnings calls.

6 Short summary

- **Conclusion.** The paper builds a new technology panel from text and uses it to show that economically important technologies are born in a few places, diffuse slowly, and remain skill biased for decades.
- **Main empirical lesson.** Technology diffusion does not quickly erase local innovation advantages. The diffusion is slow enough that the geography of invention becomes the geography of long-lived opportunity.
- **Main mechanism.** Lower-skill, use-oriented jobs spread sooner; higher-skill RDP jobs remain concentrated in pioneer locations. This is why the places that first invent technologies keep a high-skill advantage for so long.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a scalable method for identifying important new technologies from text and tracing their diffusion in jobs.
- It shows that the origins of economically important technologies are extraordinarily concentrated geographically.
- It also shows that the diffusion of technology-related employment is real but very slow, and that the most persistent part of the local advantage is concentrated in high-skill and RDP jobs.

7.2 Economic intuition

- The places that invent technologies first accumulate workers, firms, universities, and tacit know-how around those technologies.
- As technologies mature, use spreads to other places and to broader sets of workers, but frontier work stays near the original clusters longer.
- This makes technological diffusion a two-speed process: broad adoption can eventually spread, but the highest-value, highest-skill part of the ecosystem remains concentrated for much longer.

7.3 Takeaway

This paper is best understood as a measurement and stylized-facts contribution with big implications for regional development and inequality. It says that if a city or region manages to become an early pioneer in a technology that turns out to matter, the payoff can last for decades, especially through high-skill jobs. Conversely, policies that hope technology jobs will quickly spread from a few superstar places to the rest of the country are likely to be disappointed on any short or even medium horizon. The diffusion occurs, but it is slow, incomplete for a long time, and heavily tilted toward the lower-skill use of technologies rather than the frontier activities of inventing and improving them.